

Genpact

Data Engineer Interview Q&A

(2–5 YOY) – Real Mix

1 Question: Explain a real end-to-end data pipeline you built.

What they asked me

“Walk me through a data pipeline you built using Azure / AWS / GCP.”

What I said

I explained a full Azure pipeline:

Ingestion

- ADF Copy Activity for batch data
- Event-based triggers for incremental loads
- Data landed to **ADLS Gen2 (Bronze)** folder

Transformation

- Databricks notebooks running PySpark
- Cleaned & standardized → **Silver zone**
- Business transformations → **Gold zone**

Storage

- Delta Lake (ACID, schema evolution, time travel)

Serving Layer

- Synapse Serverless + Power BI
- API layer via Azure Functions for downstream usage

Tips

- Mention Bronze/Silver/Gold
- Show both tech + business understanding

2 Question: How do you handle schema drift in Databricks or ADF?

What they asked me

"Your source table added new columns. What happens, and how do you handle it?"

What I said

I followed 3 steps:

Detect schema drift

- Enable `allowSchemaDrift` in ADF
- Use schema inference in Databricks
- Compare incoming schema with target schema

Apply transformations

- Add new columns with default values
- Drop unexpected columns
- Use Delta's schema evolution:
- ```
ALTER TABLE target SET TBLPROPERTIES ('delta.allowSchemaEvolution' = true)
```

#### Update downstream layers

- Modify Silver → Gold mapping
- Update DQ rules

### Tips

- Genpact LOVES schema drift answers
- Avoid saying "Crawler fixes it automatically"

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## 3 Question: Explain your Incremental Load Strategy.

### What they asked me

“How do you perform incremental loads in ADF or Databricks?”

### What I said

I described:

#### Watermark column

- Maintain `last_modified_timestamp`
- Store watermark in Metadata table

#### Apply filter

```
SELECT *
FROM source
WHERE last_updated > @watermark
```

#### Upsert using Delta MERGE

```
MERGE INTO target t
USING updates u
ON t.id = u.id
WHEN MATCHED THEN UPDATE SET *
WHEN NOT MATCHED THEN INSERT *
```

#### Update watermark

Store new max timestamp after successful run.

#### Tips

- Genpact rejects “full load daily” answers
  - Mention “Delta MERGE + Metadata table”
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## 4 Question: PySpark Coding — Optimize a Slow Join

### What they asked me

“How will you optimize a slow join in PySpark?”

### What I said

I used 5 optimizations:

1 Broadcast small table

```
df.join(F.broadcast(dim), "id")
```

2 Repartition by join key

```
df.repartition("customer_id")
```

3 Filter early (pushdown)

4 Select necessary columns only

5 Enable AQE (Adaptive Query Execution)

```
spark.conf.set("spark.sql.adaptive.enabled", True)
```

### Tips

- Broadcast join is the magical keyword
- Genpact values Spark 3.x knowledge

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## 5 Question: SQL Scenario — Find Orders That Increased Month-over-Month

### What they asked me

“Write SQL to find customers whose monthly orders grew this month compared to last month.”

### What I said

```
WITH cte AS (
 SELECT customer_id,
 month,
 total_orders,
 LAG(total_orders) OVER (PARTITION BY customer_id ORDER BY
month) AS prev_month
 FROM monthly_orders
)
SELECT *
FROM cte
WHERE total_orders > prev_month;
```

### Tips

- Window functions = MUST
- Genpact SQL is strictly analytical

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## 6 Question: How do you solve the small file problem in Data Lake?

### What they asked me

“Your data lake has thousands of small files. How do you fix it?”

### What I said

- Use repartition

```
df.repartition(50).write.format("delta").save(path)
```

- Auto Optimize + Auto Compaction
- Delta OPTIMIZE ZORDER

```
OPTIMIZE sales ZORDER BY (date)
```

- Firehose batching (if streaming)

### **Tips**

- Mention ZORDER
  - Too many files = slow queries = real Genpact pain point
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## **7 Question: ADF Failure Debugging Scenario**

### **What they asked me**

“Copy activity fails for some files but works for others. How do you debug?”

### **What I said**

#### **Check:**

- Missing columns
- Type mismatch
- Network throttling
- Source system API limits
- Integration Runtime capacity
- File size corrupted
- Path naming issues

### **Tips**

- Structured debugging = Genpact’s favorite
  - NEVER say “restart pipeline”
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## **8 Question: Data Quality Framework in Genpact Projects**

### **What they asked me**

“What data quality checks do you apply in Silver layer?”



## **What I said**

### **DQ rules**

- Null checks
- Unique key checks
- Pattern check
- Threshold check
- Referential integrity

### **DQ Outputs**

- Valid → Silver
- Invalid → Error folder (`silver/error/table/date`)
- Summary → Log table

### **Tips**

- Mention “Error folder + Replay pipeline”
- Shows end-to-end ownership

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## **9 Question: When do you choose Parquet vs Delta vs CSV?**

### **What they asked me**

“Why did you choose Delta in your project?”

### **What I said**

#### **CSV**

- Only for raw ingestion

#### **Parquet**

- Columnar
- Compressed
- Best for analytics

## **Delta**

- ACID
- MERGE support
- Schema evolution
- Time travel
- Fast updates/deletes

## **Tips**

- Say “Delta = best choice for SCD & incremental loads”
  - Perfect Genpact answer
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# **10 Question: Data Modelling Scenario**

## **What they asked me**

“Design a model for a retail sales dashboard.”

## **What I said**

### **Fact\_Sales**

- Sales\_ID
- Product\_ID
- Customer\_ID
- Amount
- Quantity
- Timestamp

### **Dimensions**

- Dim\_Product
- Dim\_Customer
- Dim\_Date
- Dim\_Region

## **Tips**

- Genpact LOVES STAR SCHEMA
- Keep it simple + business focused